

A Survey on High-Dimensional Regression

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Abstract

This is a short abstract that summarizes the purpose of the document. You can write a concise introduction to your topic here.

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1 Introduction

1.1 Drawbacks of the Ordinary Least Square Estimator

Consider the dataset $\mathcal{D}_n := \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_i, y_i)\}$ where $(\mathbf{x}_i, y_i) \in \mathbb{R}^p \times \mathbb{R}$, we denote $\mathbf{x}_i$ as the covariates and y_i as the response. In linear regression we have the model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad (1.1)$$

where $\mathbf{y} = (y_1, \dots, y_n)^\top$, $\mathbf{X} := (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top$, $\boldsymbol{\beta}$ is the coefficient vector and $\boldsymbol{\epsilon}$ is a noise term such that $\mathbb{E}[\boldsymbol{\epsilon}] = \mathbf{0}$. Under a full rank design ($p < n$) the ordinary least square estimator of $\boldsymbol{\beta}$ is given by

$$\hat{\boldsymbol{\beta}} := (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}, \quad (1.2)$$

then the *prediction error* of the least square estimator is given by

$$\frac{1}{n} \|\mathbf{X}\boldsymbol{\beta} - \mathbf{X}\hat{\boldsymbol{\beta}}\|_2^2 = \frac{1}{n} \|\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\epsilon}\|_2^2. \quad (1.3)$$

In a Gaussian regression model, we assume $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbb{I}_n)$, and also assume a fixed-design model, then we have

$$\mathbb{E} \left[\frac{\|\mathbf{X}\boldsymbol{\beta} - \mathbf{X}\hat{\boldsymbol{\beta}}\|_2^2}{n} \right] = \frac{p}{n} \cdot \sigma^2. \quad (1.4)$$

The proof of (1.4) is given in the appendix. It suggests that the least squares estimate only provides accurate prediction in the case where $p \ll n$ or the variance of the noise σ^2 is very small (Lederer [Led22]). In a high dimensional setting where $p > n$, the prediction risk will be large and yield inaccuracies. Another way to see this is to consider the *estimation error* of $\hat{\boldsymbol{\beta}}$, which is the L^2 distance from $\boldsymbol{\beta}$ to $\hat{\boldsymbol{\beta}}$ defined as $L = \|\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}\|_2$, then we have

$$\mathbb{E}[L^2] = \mathbb{E}[(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})] = \sigma^2 \cdot \text{trace}(\mathbf{X}^\top \mathbf{X})^{-1} \quad (1.5)$$

and under a Gaussian regression model $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbb{I}_n)$ we have

$$\mathbb{V}(L^2) = 2\sigma^4 \cdot \text{trace}(\mathbf{X}^\top \mathbf{X})^{-2}. \quad (1.6)$$

The proof of above two equations is given in the appendix. Hence the least squares estimate will again yield inaccuracies if $\mathbf{X}^\top \mathbf{X}$ moves from a unit matrix to a ill-conditioned one with small eigenvalues (Hoerl and Kennard [HK70]).

1.2 High Dimensional Data and Simulations

The words “high dimensional” refers to the situations where the number of covariates is larger than the number of data points, i.e., $p > n$. Such situations happen in many fields nowadays, including genomics, text classification, image analysis, astronomy, atmospheric science, etc. (Siri-mongkolkasem and Drikvandi [SD19]; Johnstone and Titterton [JT09]). In such case the design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ is not full (column) rank and hence the least squares estimate (1.2) will yield infinitely many solutions, leading to overfitting. Regularization methods such as Ridge regression (Hoerl and Kennard [HK70]), Lasso (Tibshirani [Tib96]), Elastic Net (Zhou and Hastie [ZH05]),

Bayesian Lasso (Park and Casella [PC08]), and many other were proposed for analysing high-dimensional data by imposing a penalty on the number of covariates. In this section we examine several real data sets and computer simulations with the case $p > n$ and we compare the performances of different regularization methods. We formally study those regularization methods in section 2.

Example 1: A Computer Simulation: We consider a Gaussian regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbb{I}) \quad (1.7)$$

with $p = 20$ and $n = 10$. For each $x_{ij} \in \mathbf{X}$ ($i \in [10], j \in [20]$), we set $x_{ij} \sim \mathcal{N}(0, (0.5)^{|i-j|})$ and the true coefficients vector as $\boldsymbol{\beta} = (1.5 \ 2 \ 3 \ 2.5 \ 1.8 \ 0 \ \cdots \ 0)$. We apply 3 different regularization methods: Ridge, Lasso, Elastic Net. Their corresponding estimates are

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} := \arg \min_{\boldsymbol{\beta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right\} \quad (1.8)$$

$$\hat{\boldsymbol{\beta}}_{\text{Lasso}} := \arg \min_{\boldsymbol{\beta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\} \quad (1.9)$$

$$\hat{\boldsymbol{\beta}}_{\text{ENet}} := \arg \min_{\boldsymbol{\beta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2 \right\} \quad (1.10)$$

where the tuning parameters $\lambda, \lambda_1, \lambda_2$ are selected using Leave-One-Out Cross-Validation (LOOCV). The performance is shown in figure 1 and table 1.

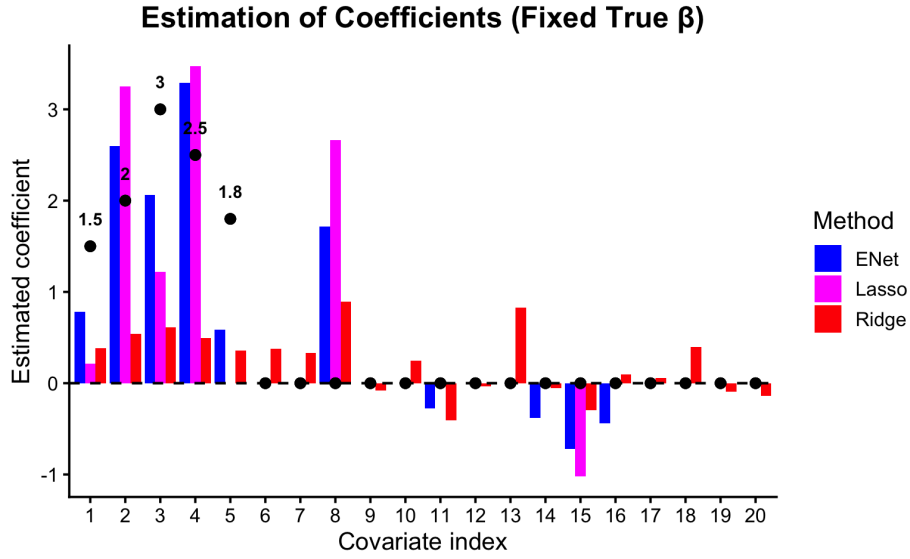


Figure 1: The estimation of each coefficient by different regularization methods where the black dots indicate the true value.

Method	Estimation Error	Prediction Error	Sparsity Index
Lasso	2.1835	1.0921	5
Ridge	3.9478	1.3586	20
Elastic Net	2.5456	1.1034	6

2 Regularization and Variable Selection

2.1 Ridge Regression

In order to control the inflation and instability associated with the least squares estimates, Horel ([[HK70](#)]) suggests the following form

$$\hat{\boldsymbol{\beta}}^* := (\mathbf{X}^\top \mathbf{X} + k\mathbb{I})^{-1} \mathbf{X}^\top \mathbf{y} \quad k \geq 0 \quad (2.1)$$

3 Appendix

Proposition 1. Consider a fixed design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ and a Gaussian regression model where $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbb{I}_n)$, let $\hat{\beta}$ be the least squares estimate of β , then

$$\mathbb{E} \left[\frac{\|\mathbf{X}\beta - \mathbf{X}\hat{\beta}\|_2^2}{n} \right] = \frac{p}{n} \cdot \sigma^2 \quad (3.1)$$

Proof. By using $\mathbf{y} = \mathbf{X}\beta + \epsilon$ and $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, it is easy to see that

$$\mathbb{E} \left[\frac{\|\mathbf{X}\beta - \mathbf{X}\hat{\beta}\|_2^2}{n} \right] = \frac{1}{n} \mathbb{E} \left[\|\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon\|_2^2 \right]. \quad (3.2)$$

The square of the L^2 norm is given by

$$\begin{aligned} \|\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon\|_2^2 &= \left(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon \right)^\top \cdot \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon \\ &= \epsilon^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X} \epsilon \end{aligned} \quad (3.3)$$

Define $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}$ as the hat matrix, then we have

$$\begin{aligned} \frac{1}{n} \mathbb{E} [\epsilon^\top \mathbf{H} \epsilon] &= \frac{1}{n} \mathbb{E} [\text{trace}(\epsilon^\top \mathbf{H} \epsilon)] \\ &= \frac{1}{n} \mathbb{E} [\text{trace}(\mathbf{H} \epsilon \epsilon^\top)] \\ &= \frac{1}{n} \text{trace}(\mathbb{E}[\mathbf{H} \epsilon \epsilon^\top]) \\ &= \frac{1}{n} \text{trace}(\mathbf{H} \cdot \sigma^2) \\ &= \frac{p}{n} \cdot \sigma^2. \end{aligned} \quad (3.4)$$

The property of hat matrix has that $\text{trace}(\mathbf{H}) = p$. □

Proposition 2. Consider a Gaussian regression model, define $L = \|\beta - \hat{\beta}\|_2^2$, then

$$\mathbb{E}[L^2] = \sigma^2 \cdot \text{trace}(\mathbf{X}^\top \mathbf{X})^{-1} \quad \mathbb{V}(L^2) = 2\sigma^4 \cdot \text{trace}(\mathbf{X}^\top \mathbf{X})^{-2}. \quad (3.5)$$

Proof. By the property of trace operator we have

$$\mathbb{E}[L^2] = \mathbb{E}[\text{trace}(\beta - \hat{\beta})^\top (\beta - \hat{\beta})] = \text{trace}(\mathbb{E}[(\beta - \hat{\beta}) \cdot (\beta - \hat{\beta})^\top]). \quad (3.6)$$

By the fact that $\hat{\beta}$ is unbiased of β , we have

$$\text{trace}(\mathbb{E}[(\beta - \hat{\beta}) \cdot (\beta - \hat{\beta})^\top]) := \text{trace}(\mathbb{V}(\hat{\beta})) := \sigma^2 \cdot \text{trace}(\mathbf{X}^\top \mathbf{X})^{-1}. \quad (3.7)$$

As for $\mathbb{V}(L^2)$, observe that $\beta - \hat{\beta} = -(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon$, hence

$$(\beta - \hat{\beta})^\top \cdot (\beta - \hat{\beta}) = \epsilon^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-2} \mathbf{X}^\top \epsilon := \epsilon^\top \mathbf{\Lambda} \epsilon. \quad (3.8)$$

where $\mathbf{\Lambda} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-2} \mathbf{X}^\top$. Denote $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_n)^\top$ and λ_{ij} as the $i - j$ element in $\mathbf{\Lambda}$, we have

$$(\boldsymbol{\epsilon}^\top \mathbf{\Lambda} \boldsymbol{\epsilon})^2 = \left(\sum_{i,j} \epsilon_i \lambda_{ij} \epsilon_j \right)^2 = \sum_{i,j,k,l} \epsilon_i \lambda_{ij} \epsilon_j \epsilon_k \lambda_{kl} \epsilon_l = \sum_{i,j,k,l} a_{ij} a_{kl} \epsilon_i \epsilon_j \epsilon_k \epsilon_l. \quad (3.9)$$

In a Gaussian regression model, we have that

$$\mathbb{E}[\epsilon_i \epsilon_j \epsilon_k \epsilon_l] = \sigma^4 (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \quad \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (3.10)$$

so we have

$$\begin{aligned} \mathbb{E} \left[\sum_{i,j,k,l} \lambda_{ij} \lambda_{kl} \epsilon_i \epsilon_j \epsilon_k \epsilon_l \right] &= \mathbb{E} \left[\sum_{i,j} \lambda_{ii} \lambda_{jj} \epsilon_i^2 \epsilon_j^2 + 2 \sum_{i,j} \lambda_{ij} \lambda_{ji} \epsilon_i^2 \epsilon_j^2 \right] \\ &= \sigma^4 \cdot \mathbb{E} \left[(\text{trace}(\mathbf{\Lambda}))^2 + 2(\text{trace}(\mathbf{\Lambda}^2)) \right]. \end{aligned} \quad (3.11)$$

Note that we also have

$$\text{trace}(\mathbf{\Lambda}) = \text{trace}(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-2} \mathbf{X}^\top) = \text{trace}(\mathbf{X}^\top \mathbf{X})^{-1}, \quad (3.12)$$

$$\text{trace}(\mathbf{\Lambda}^2) = \text{trace}((\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-2} \mathbf{X}^\top)^2) = \text{trace}(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-3} \mathbf{X}^\top) = \text{trace}(\mathbf{X}^\top \mathbf{X})^{-2}, \quad (3.13)$$

so the variance of L^2 is given by

$$\begin{aligned} \mathbb{V}(L^2) &= \mathbb{E}[L^4] - (\mathbb{E}[L^2])^2 \\ &= \sigma^4 \cdot \mathbb{E} \left[\left(\text{trace}(\mathbf{X}^\top \mathbf{X})^{-1} \right)^2 + 2 \text{trace}(\mathbf{X}^\top \mathbf{X})^{-2} \right] - \mathbb{E} \left[\left(\sigma^2 \cdot \text{trace}(\mathbf{X}^\top \mathbf{X})^{-1} \right)^2 \right] \\ &= 2\sigma^4 \cdot \text{trace}(\mathbf{X}^\top \mathbf{X})^{-2}. \end{aligned} \quad (3.14)$$

□

4 References

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